

Summary of Logistic Regression Model Performance

The Logistic Regression model shows the following performance:

Precision, Recall, F1-score:

Negative: Precision: 1.00, Recall: 0.33, F1-score: 0.50

Neutral: Precision: 0.49, Recall: 0.53, F1-score: 0.51

Positive: Precision: 0.94, Recall: 0.93, F1-score: 0.94

Overall Accuracy: 0.88

Interpretation of Confusion Matrix:

The confusion matrix for Logistic Regression (shown above) indicates:

The model correctly identified 1 out of 3 'Negative' reviews, but misclassified 2 as 'Neutral'.

For 'Neutral' reviews, 18 were correctly classified, while 16 were misclassified as 'Positive'.

The model performed very well on 'Positive' reviews, correctly classifying 239 out of 256, with 17 misclassified as 'Neutral'.

This suggests that the Logistic Regression model struggles with the 'Negative' and 'Neutral' classes, likely due to the class imbalance, but performs strongly on the majority 'Positive' class.